`ComplexGitSync` 8.4.0– Getting Started

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1 Getting Started

This chapter takes a new user from installation to a first persisted State of a project. It follows the structure of the project's `README.md`, with the detail needed to act on each part. Every command and option is described in full in the User Guide.

2 What `ComplexGitSync` is for

Scientific projects are becoming more complex: beyond source code, they rely on documentation, tutorials, data, continuous integration and deployment, and the persistence of their results — each often held in a Git repository of its own. `ComplexGitSync` manages such a project as a distributed, git-native operating space, in the form of a **GitTree**: a tree of nested Git repositories that is synchronised, versioned and restored as one unit.

A project can couple **public and private repositories**, so that intellectual property is preserved even when the project is developed collaboratively. Private repositories can also carry the configuration of **agentic assistance** for different parts of the project. Every state the project goes through is recorded, which gives the project **persistence and reproducibility**: any recorded state can be rebuilt, exactly, on another machine.

3 The project lifecycle

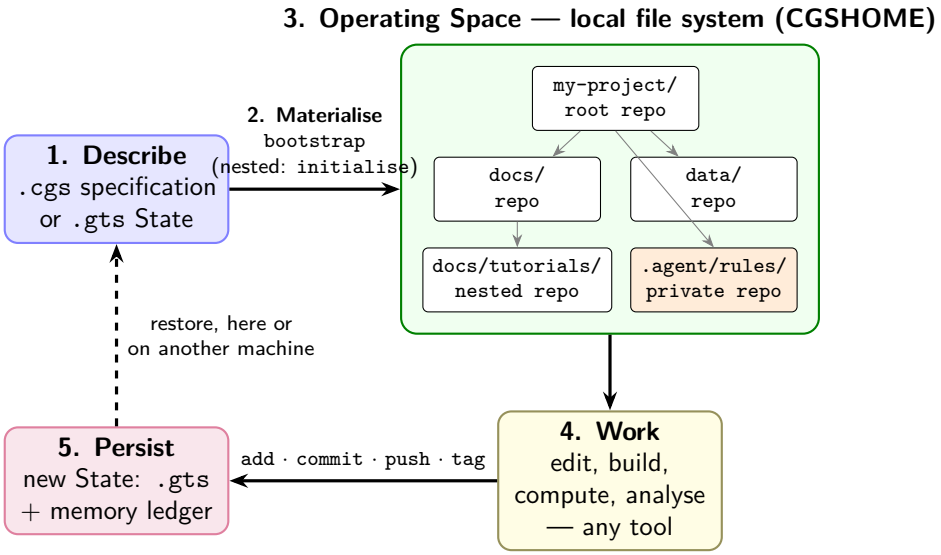


Figure 1: The lifecycle of a project: a description is materialised as a local file system, the Operating Space; the project is worked on with any tool; its new state is persisted, and can be restored from there.

Figure 1 shows the five steps every project goes through. Each is described below with the vocabulary the rest of this manual uses.

1. Describe. The project's topology is written once in a `.cgs` specification, a short TOML file: which repositories the project is made of, where each one sits in the tree, and which are private. A `.cgs` names branches, so it always gives the latest work. A `.gts` snapshot recorded earlier can be used in its place: it names exact commits, so it gives the project exactly as it was.

```
project = "my-project"
repos = [
  "github:me/my-project",           # the root repository
  "github:me/my-project-docs",      # a project repository
  { repository = "github:me/.myRules",
    private = true, writable = true },   # a private one
]
```

Each identifier has the form `provider:owner/name`; the providers are `github`, `gitlab`, `codeberg` and `custom`. Branch, protocol and path all have defaults, and the file `examples/template.cgs` is a starting point. Before anything is cloned, `validate` checks a `.cgs` and `view-tree` draws the tree it describes:

```
$ pixi run cgitsync validate my-project.cgs
$ pixi run cgitsync view-tree my-project.cgs
```

2. Materialise. `bootstrap`, the generic command for any kind of project, clones every repository the description names and places it at its path in the tree. `initialise` is its nested counterpart, run from inside the project (Section 7).

3. The Operating Space. The result is an ordinary directory tree on the local disk, called the project's `CGSHOME`. Some of its sub-directories are Git repositories of their own, public or

private, nested at the paths the description gave them. ComplexGitSync keeps its own records in `CGSHOME/.cgitsync/`: the States, the ledger, and a log of every command it ran.

4. Work. The project files are used with any tool, exactly as in any other directory: editors, compilers, notebooks, simulation codes, plain `git` inside one repository. ComplexGitSync does not intervene.

5. Persist. `cgitsync` runs each Git operation across the whole tree, in the right order, and checks every repository before changing any of them. Each command that changes the tree records the resulting **State**: a `.gts` snapshot holding the exact commit of every repository, entered in the project's **memory**, a hash-chained ledger that can be verified. A State can be given to `bootstrap` (or, in the nested mode, to `initialise`) to restore it — which closes the cycle.

4 Public and private repositories

A GitTree holds two kinds of repositories.

Project repositories are the work itself: code, documents, public data. They all follow the project's branch: `cgitsync checkout feature` moves every one of them to `feature`.

Private repositories are how the project is run: pipelines, agent instructions, internal rules, private data. They are declared with `private = true`, are often shared between several projects, and keep their own branch rather than following the project's.

A private repository is read-only unless its entry adds `writable = true`. A writable one is written to only on request: commands act on the project repositories by default, and on the writable private ones with `-private`. ComplexGitSync never writes to a read-only one.

```
$ pixi run cgitsync commit "update the analysis"            # project
    repositories
$ pixi run cgitsync commit --private "tighten a rule"       # writable
    private ones
```

`tutorials/05_private_repos.md` works through a complete example.

5 Memory and persistence

Every State, and every operation that produced one, is entered in the project's memory, together with the tool versions and the environment it ran in. A project with no private repository keeps its memory on the local disk only. A project with private repositories is expected to keep it in a private repository of its own, so that its history outlives any single machine; `cgitsync` offers to set this up the first time it records something:

```
$ pixi run cgitsync memory setup       # create and adopt the memory
    repository
$ pixi run cgitsync memory list        # every recorded State, newest
    first
$ pixi run cgitsync memory as-of 2026-09-01  # the project on that date
$ pixi run cgitsync verify check       # does the recorded history verify
    ?
```

Once adopted, the memory is sent with every `push`, `tag` and `release freeze`. `tutorials/06_memory.md` covers it in full.

6 Installation

Tool	Why it is needed
Git	Clones and synchronises every repository.
Pixi	Installs the locked Python environment and runs `cgitsync`; Python itself comes with it.
`gh`, `glab` or `tea`	Optional: creates repositories on GitHub, GitLab or Codeberg (`repo create`, `memory setup`).

Git authentication must also work for every remote the project uses.

```
$ git clone https://github.com/flipoyo/ComplexGitSync.git
$ cd ComplexGitSync
$ pixi install
$ pixi run cgitsync --help
```

`cgitsync` is a Pixi task, not a globally installed program: every command is typed as `pixi run cgitsync ...` from inside the ComplexGitSync clone.

7 Standalone or nested

The ComplexGitSync clone can manage a project from outside it or from inside it. The choice is made once, by the command used to materialise the project.

Standalone (recommended). ComplexGitSync stays where it was cloned, and `bootstrap` clones the whole project tree, root included, into a new workspace of its own, by default `$HOME/.cgs/<project name>-<timestamp>`:

```
$ pixi run cgitsync bootstrap my-project.cgs
$ export CGSHOME=/home/me/.cgs/my-project-<timestamp>   # printed by
    bootstrap
$ pixi run cgitsync status
```

Since commands are run from the ComplexGitSync clone, the `CGSHOME` variable tells them which workspace to act on. Every command prints the workspace it picked and why — from `$CGSHOME`, or from the current directory — and warns when that is not where you are standing — the usual sign of an export left over from an earlier project.

Nested. The project's root repository is cloned first, and ComplexGitSync is cloned inside it; `initialise` then builds the rest of the tree around them, with no export needed. This suits projects that ship the tool along with their own content, such as hydrological digital twins, where whoever obtains the project obtains the means to synchronise it:

```
$ git clone https://gitlab.com/CGS_test/CGSil1.git
$ cd CGSil1
$ git clone https://github.com/flipoyo/ComplexGitSync.git
$ cd ComplexGitSync
$ pixi install
$ pixi run cgitsync initialise ../CGSil1.cgs --force-protocol https
```

Each command refuses the other's job by name before touching the disk: `initialise` run from outside a checked-out root names `bootstrap`, and `bootstrap` into a directory that already holds a checkout names `initialise`. `status` reports which case is in force, as `use_case=standalone` or as `use_case=nested`.

On Windows. The command line is Linux-like and the tutorials are written for bash, which on Windows is Git Bash (it comes with Git for Windows). The lines to paste are printed with forward slashes (`export CGSHOME=C:/Users/you/.cgs/my-project-<timestamp>`), so they work as printed and no backslash is ever doubled; other messages may still show a path with backslashes. `~` and `$HOME` name the same directory to the shell and to the tool, and the default workspace root is `$HOME/.cgs`; only when `HOME` is unset does the tool use the profile directory. In another shell, only the lines that are shell syntax change: `$env:CGSHOME = "C:/Users/you/.cgs/my-project-<timestamp>"` in PowerShell, `set CGSHOME=C:/Users/you/.cgs/my-proj` in `cmd`, and `curl.exe` for PowerShell's `curl` alias; `tutorials/README.md` has the table.

SSH or HTTPS. A `.cgs` clones over SSH by default, and that stays the default for your own projects. Without an SSH key on GitLab or GitHub (`ssh -T git@gitlab.com` tells you), add `-force-protocol https` to the command that clones, as above: the tutorials do it on every such line, because their reader usually has no key yet. To push over HTTPS, let the provider's tool give Git your credentials once (`glab auth login`; `gh auth login` then `gh auth setup-git`); `cgitsync` never reads a token. Once you have a key, `pixi run cgitsync pull -force-protocol ssh` switches the whole tree to SSH and changes no State. The memory's repository is always reached over SSH. Every detail is in Chapter **??**, Section **??**.

8 A first project

Choose the starting point that matches the project; each has a tutorial that walks through it end to end.

To discover the tool on a sample tree. The small public tree `CGSil1` has nothing of your own at risk. `tutorials/01_first_multi_repo_workspace.md` installs it with `bootstrap`, and shows the same tree installed nested with `initialise` so the difference is concrete; `tutorials/02_working_with_a_tree.md` then runs the whole lifecycle on it, on a branch of its own.

The project already provides a `.cgs`. This is the usual case, for any kind of project: use `bootstrap`, as in Section 7. `tutorials/03_onboarding_a_real_build_tree.md` does this for a real 19-repository project.

The project is on disk but has no `.cgs`. `discover` scans a directory for Git repositories and drafts the `.cgs` from what is checked out. It writes nothing until `-write` is given, and a relative file name is written inside the scanned directory; review the draft before using it.

```
$ pixi run cgitsync discover ~/work/my-project --write my-project.cgs
$ pixi run cgitsync validate ~/work/my-project/my-project.cgs
```

The project uses git submodules. `submodules init` adopts the checkout in one command — discover, write the `.cgs`, initialise, then turn each submodule into a plain nested repository — and `-dry-run` shows the plan first. It leaves the conversion staged, not committed. `tutorials/04_adopting_a_real_project.md` walks through it.

```
$ pixi run cgitsync submodules init ~/work/my-project --dry-run
$ pixi run cgitsync submodules init ~/work/my-project
```

9 Day-to-day work

Once the tree is materialised, each Git operation is run once, for the whole tree. `tutorials/02_working_with_a_tree.md` works through these commands on the sample tree, including going back to a release from its State:

```
$ pixi run cgitsync status                  # where every repository
    stands
$ pixi run cgitsync view-tree               # the tree, drawn
$ pixi run cgitsync pull                    # bring every repository up
    to date
$ pixi run cgitsync branch create my-feature   # a branch in every
    repository
$ pixi run cgitsync branch list             # the project's branches, *
    on the current one
$ pixi run cgitsync checkout my-feature     # move the project to a
    branch
  ... work on the files ...
$ pixi run cgitsync add
$ pixi run cgitsync commit "what changed"
$ pixi run cgitsync push
$ pixi run cgitsync release freeze "first release"   # add, commit, pull
    , push, tag, record
$ pixi run cgitsync release list                     # every release,
    whoever made it
$ pixi run cgitsync release load 1                   # back to a release
    , detached
```

`status` prints one row per repository — its branch, whether its files are `clean` or `dirty`, and whether it is `synced`, `ahead` or `behind` its remote — and a summary line naming the project's branch. `pull` visits the tree from the root down; `add`, `commit` and `push` from the leaves up, so a parent never records a child that is not yet committed. `add`, `commit`, `push` and `release freeze` accept `-dry-run` to show what they would do.

Commands that could lose work refuse instead. `merge` checks every repository before merging any, so a conflict anywhere leaves the whole tree untouched; `pull -force` refuses while a commit exists that no remote holds, and sets uncommitted work aside with `git stash` rather than discarding it.

10 Getting help

```
$ pixi run cgitsync --help              # every command, grouped by
    purpose
$ pixi run cgitsync <command> --help    # one command: purpose, options,
    examples
$ pixi run cgitsync help --all          # every command and option on one
     page
```

`-h` is accepted wherever `-help` is. When a command fails, it prints one line saying why, and its full log is kept in `CGSHOME/.cgitsync/logs/`; `cgitsync autofix` reads the last one and diagnoses the situation.

11 Next steps

The eight tutorials under `tutorials/` work through the cases above, from the simplest to the most advanced. The User Guide is the reference for every command, option, document format

and error.